



Kulpio

Waste less food. Save money. In any language.

A single-file offline PWA + one Cloudflare Worker that helps households waste less food — AI input, trustworthy expiry, a shared fridge, 33 languages.

INFOMATRIX 2026

PROGRAMMING

Daniil Bejenari

T.L. Orizont · Ceadâr-Lunga

THE PROBLEM

A third of our food is wasted — mostly at home

~1/3

of all food produced is lost or wasted (FAO)

#1

households are among the largest single contributors

Forget

the main cause at home:
bought, forgotten, expired



It's an information & attention problem — not a preservation problem.
That's exactly what an app can fix.

Sources: FAO — Global Food Losses & Food Waste · UNEP — Food Waste Index Report.

Built for the Sustainable Development Goals



PRIMARY

Responsible Consumption & Production

Kulpio's core mission — cut the food households throw away.



Zero Hunger

Use more of the food we buy; value it instead of binning it.



Climate Action

Less wasted food means fewer needless food-related emissions.



UN Target 12.3: halve per-capita global food waste by 2030. Kulpio goes straight at household waste — the largest, most-forgettable slice.

Track it, know what's going off, cook it first

01



Add in seconds

Scan a barcode, read a label or receipt with AI, or paste a list — no typing marathon.

02



Know what expires

Printed dates beat estimates. Live freshness colours and a 'week-ahead' strip.

03

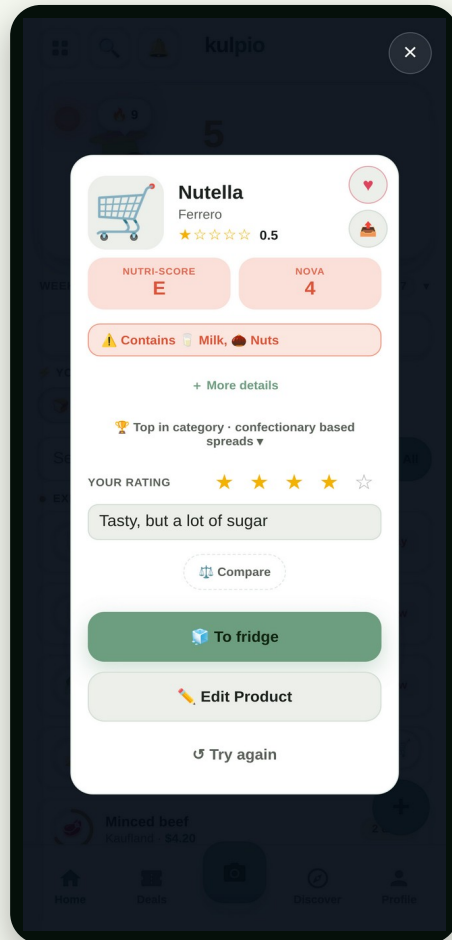


Use it in time

Cook what's going off first; mark Used / Wasted / Froze and watch the money you save.

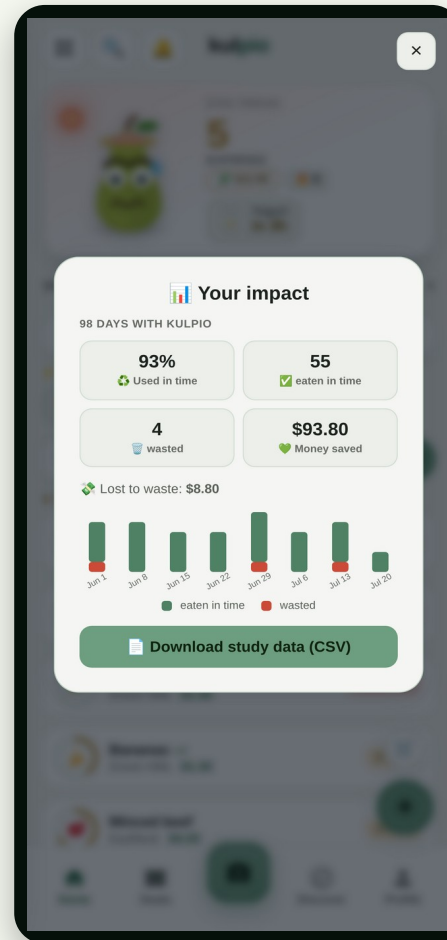
SEE IT

One app, offline, on any phone



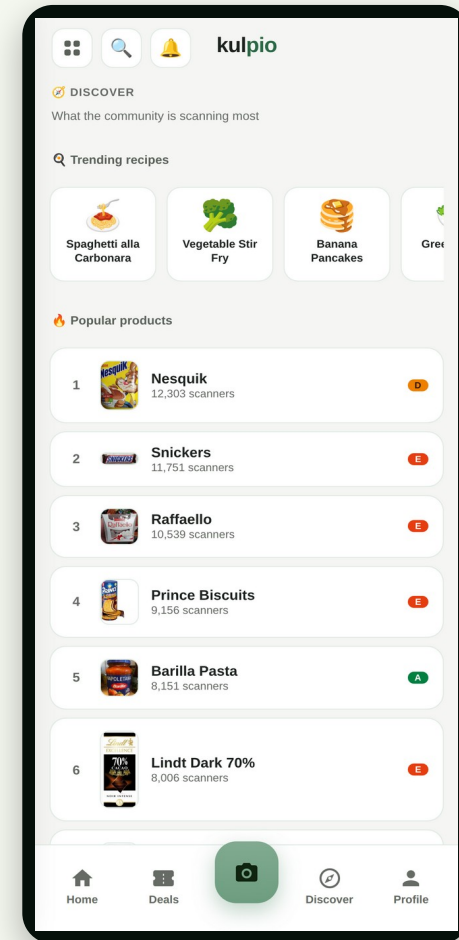
Scan → product card

Kulpio · food-freshness tracker
Nutri-Score, allergens, verdict



Your impact

waste cut & money saved

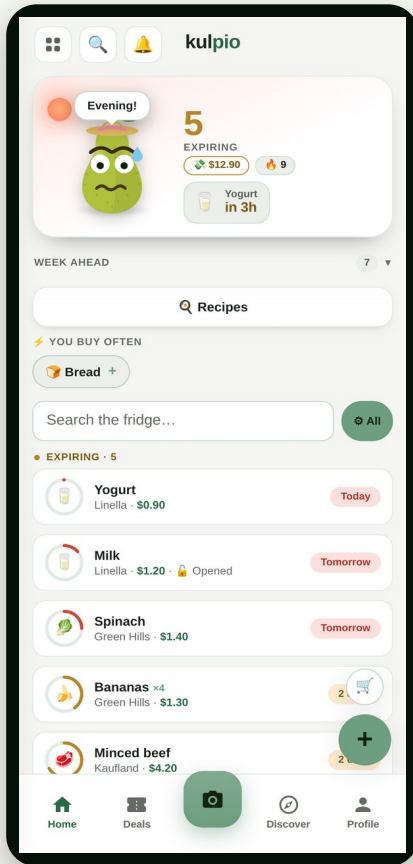


Discover

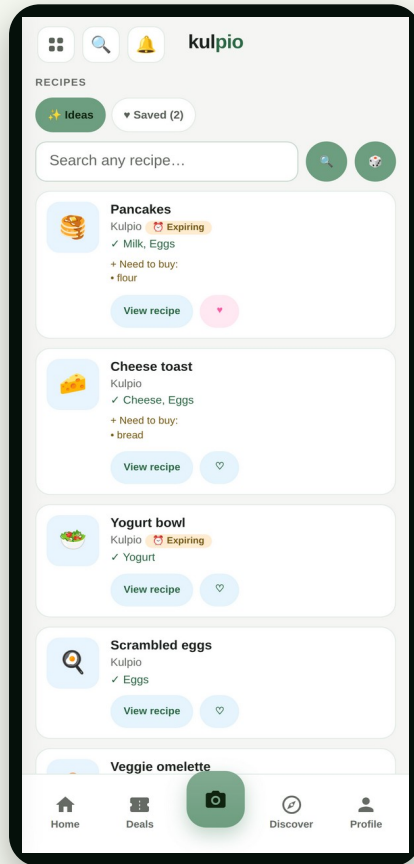
what the community scans

A CLOSER LOOK

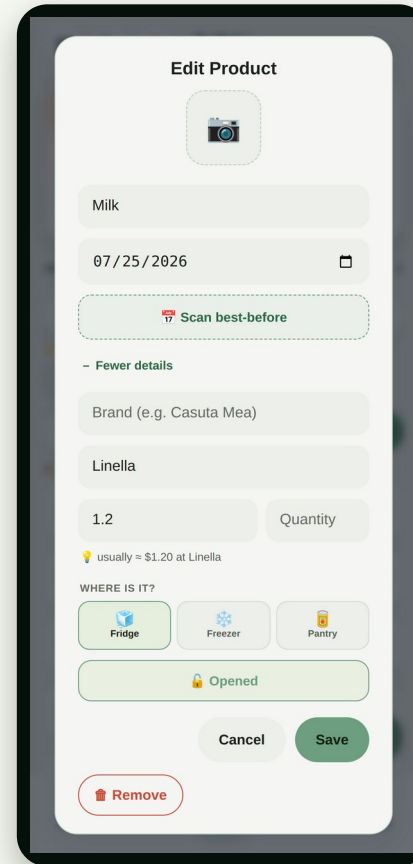
A quick tour of the app



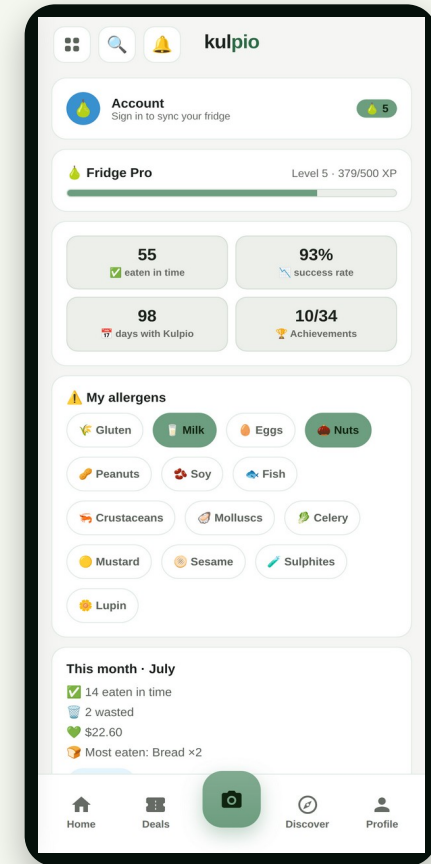
Home dashboard
today, week-ahead & fridge



Recipes
cook from what you have



Every item, one tap
edit, price history, storage



Profile & levels
achievements, XP & impact

What Kulpio does



Effortless input

Barcode, AI label & receipt scan,
or paste a list



Storage sections

Fridge, freezer & pantry with live
freshness



Recipes

Cook from your ingredients, with
nutrition estimates



Shared fridge

One code links a household +
activity feed & chat



Motivation

Achievements, XP levels, monthly
recap & impact



33 languages

Full RTL support; installs to the
home screen

How Kulpio decides when food expires

Printed dates are ground truth — Kulpio only estimates when there's no date to read.

1

Printed date wins

Read from the pack (OCR + model) or typed by you. Always takes priority.

2

Offline shelf-life table

Built-in per-food estimate, adjusted for storage (freezer +~90 d; opened shortens).

3

AI refine

When online, the model refines the estimate from the product name.

AI degrades gracefully

Best cloud AI model (top quality)



Cloudflare Workers AI — Llama (free)



Clean error — never blocks

Built as one offline file + one worker



Client

- **Single static HTML** — ~1.3 MB — markup, CSS, JS, 33 langs
- **No build step** — no framework, no bundler
- **Service worker** — precache → fully offline
- **Web Crypto** — hashing & secure tokens



Backend

- **Cloudflare Worker** — serves the app AND the API
- **D1 (SQLite)** — accounts, sync, households, community
- **Web Push (VAPID)** — daily expiry reminders
- **Rate-limited auth** — PBKDF2 · 5 tries / 15 min

Deploy: one Cloudflare Worker serves static assets + API on a single origin — installs from static hosting, runs offline.

Why Kulpio is different

	Offline, no install	AI label/receipt	Shared fridge + chat	33 languages	Privacy-first
Kulpio	✓	✓	✓	✓	✓
Inventory apps (NoWaste...)	✗	partial	✗	few	varies
Too Good To Go	✗	✗	✗	some	account
Whisk / Samsung Food	✗	partial	✗	some	cloud-first

The combination — *offline single-file PWA + AI input + a social shared fridge + privacy* — is the edge.

Private by design, hardened by default



Local-first

Your fridge stays in the browser — nothing leaves unless you choose a networked feature.



GDPR export

One tap downloads everything the app holds about you as JSON.



Real deletion

Delete account wipes the server account, sessions & synced data.



Hardened auth

PBKDF2 hashing, opaque tokens, and login rate-limiting (5 / 15 min).

Built to last — and tested

1

static HTML file — no build step, no framework

33

languages, with full right-to-left support

300+

automated checks run offline on every push

4

test suites: structure, smoke, worker auth & sync

Every push runs the full suite in CI — text guard-rails, a 300+-check headless Playwright smoke test that runs the app fully offline, and Node tests for auth, push/cron and household-merge.

AI degrades gracefully: best cloud model → free Cloudflare Workers AI (Llama) → clean error. The app never blocks on AI.

How it evolved — and what I had to solve

01

Offline core

Single-file PWA + shelf-life table.

Challenge solved

Had to run with no network → service-worker precache.

02

Trustworthy dates

AI label reading + estimates.

Challenge solved

Estimates felt unreliable → printed dates always win.

03

Kill the friction

Barcode, receipt scan, paste-a-list.

Challenge solved

Typing was tedious → four fast input paths.

04

Social & safe

Shared fridge, accounts, GDPR, 33 langs.

Challenge solved

Two phones clobbered data → server-side merge.

Where Kulpio goes from here



Roadmap

- Native Android app on the Play Store
- Run the real user pilot & publish measured results
- Validate the offline shelf-life table against reference data
- Email verification (add an email provider)
- Field-level merge for multi-device offline edits
- A shareable monthly impact report



Free & open-source

Released under the GNU AGPL v3 — free for anyone to use, study and build on.

The vision: help households everywhere waste less food and save money — in their own language.



Waste less. Save more. Together.

Kulpio — a food-freshness tracker that fits in your pocket, in your language.

SDG 12

SDG 2

SDG 13

Daniil Bejenari · T.L. Orizont · Ceadîr-Lunga

github.com/Gugster777/kulpio · Infomatrix 2026